UNIVERSITY

Fatigue, Recover, Adapt

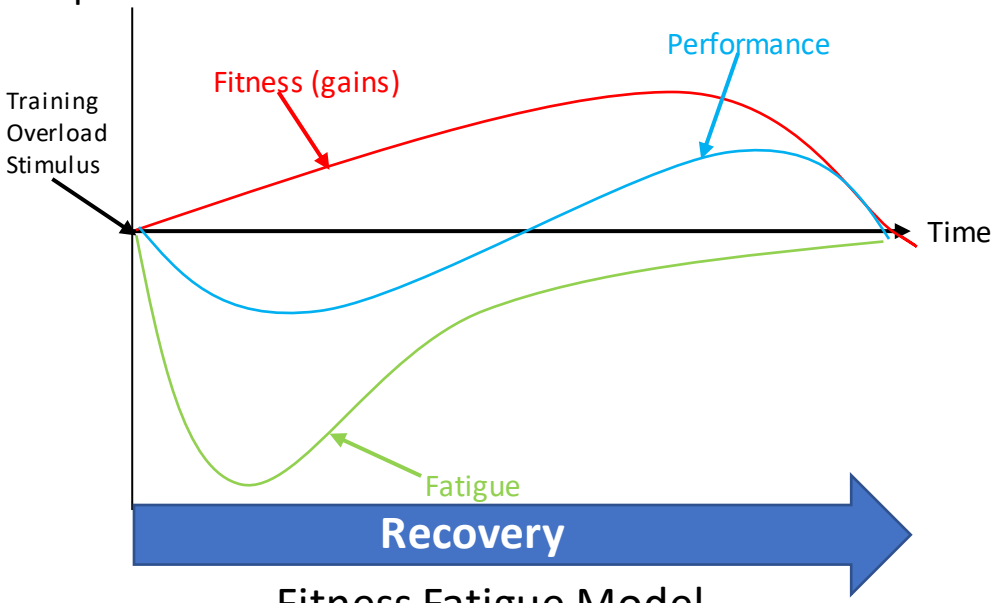
Recover from training to optimize adaptation

The Lesson

- Understand fatigue and the causes
- Fatigues effect on performance and body composition
- Detecting fatigue
- Understand recovery and adaptation
- Recovery strategy overview

Definitions

- Overload: Acute disruption to homeostasis ie resistance training stress
- Fatigue: reduction in the capacity to produce performance ie strength (fitness)
- Recovery: return to previous performance levels or better prior to disruption
- Adaptation: Process leading to functional improvement in performance



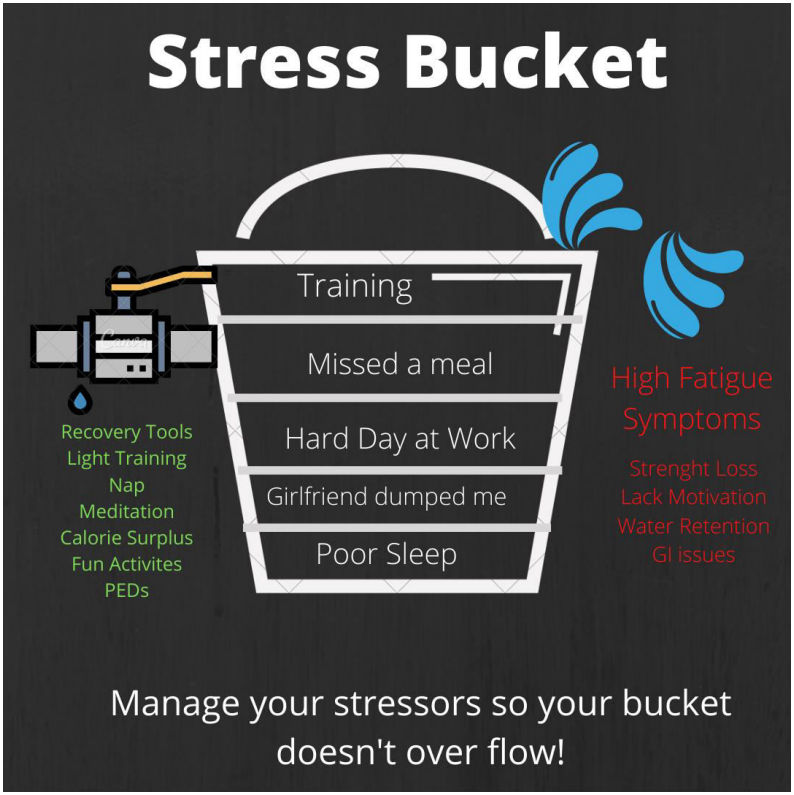
Fitness Fatigue Model

Recover and Adapt

- An overload stimulus is needed to cause an adaptation, but recovery dictates the degree of that adaptation.
- Poor overload, no adaptation, little recovery needed
- Too much overload, adaptation occurs, long recovery time limits next overload session, adaptation diminishes
- You need the right amount of overload to adapt and optimize recovery to go train and overload again. (MORE YOU DO THIS FASTER YOU MAKE GAINS)

Causes of Fatigue

- Training
  - Volume
  - Absolute and Relative Intensity
  - Frequency
  - Type (plyo, cardio, resistance) and Muscle Damage
- Life Stressor
  - Daily Activity
  - Poor Sleep
  - Poor Nutrition
  - Mental Stress
  - Illness
  - Drugs/Alcohol



Fatigue Types

- Acute Fatigue: fatigue effecting you during and immediate after training
  - Acute Fatigue
  - ATP depletion
  - Creatine Depletion
  - Nervous System Disruption
  - Oxygen Depletion
  - Decrease Blood Sugar
  - Metabolite Increase
- Chronic (cumulative) Fatigue; This is longer lasting and can accumulate over days and weeks.
  - Glycogen Depletion
  - CNS disruption
  - Endocrine, Paracrine, Autocrine Disruption (testosterone, cortisol)
  - Psychological Stress
  - Tissue Trauma
- Peripheral (Local) Fatigue: Fatigue localized to the muscle and tissues at work
- Systemic (Central) Fatigue: culmination of all fatigue in the body (peripheral, mental stress, etc)

How Does Fatigue Effect Training

- Decrease in speed and power
- Decrease in learning new lifts
- Decreased strength
- Decrease in Anabolic Process (drop in Mtor)
- Increase in Catabolic Processes (increase in AMPK)
- Body Composition (muscle loss, fat gain. Difficulty in fat loss)
- Training stimulates less protein synthesis
- Decreased immunity
- Increased injury risk

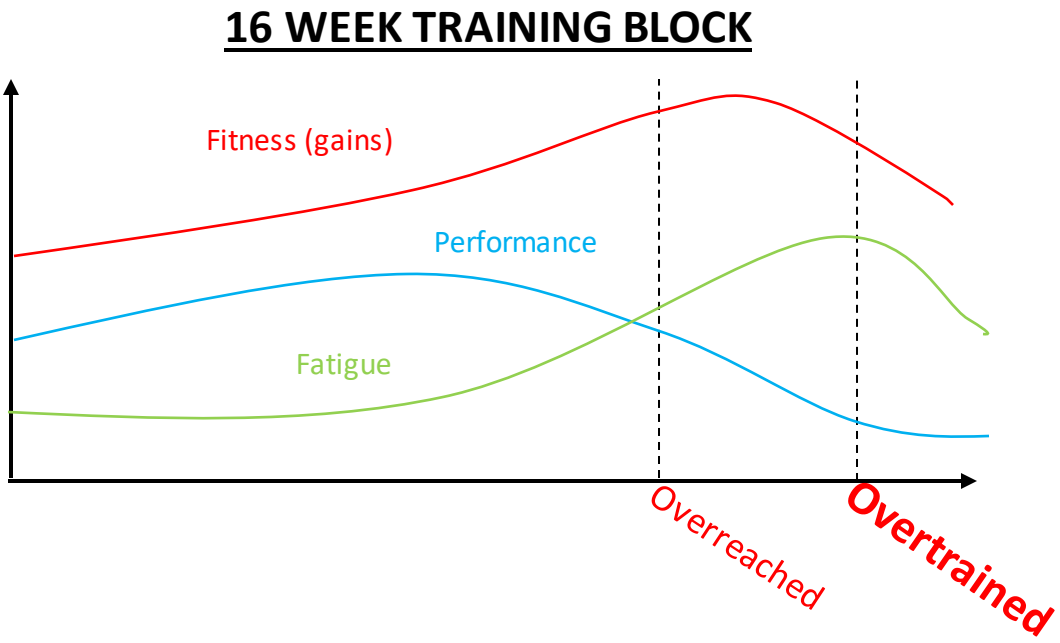
Signs of High Fatigue

Rate of Perceived Exertion/Stress
Gym Performance
Strength
Decreased Muscle Pump
Heart Rate Variability
Training Motivation
Mood Changes
Appetite Suppression
GI disruption
Sleep Disruption
Illness
Injuries and Stiffness
Loss of Libido and Menses

Overreaching and Overtraining

Overreaching: Temporary decline in training performance, can take a few days to reverse

Overtraining: Chronic decline in training performance, take weeks to months to reverse.



Assessing Fatigue

- Perceived Recovery Scale 1-10
- Sleep Quality 1-10
- Log Book: Strength Decrease, Increase, Maintain?
- Life Stress 1-10
- Aches and Pains Increase Y or N?
- Are you Motivated to train? Y or N
- Rate Hunger 1-10?
- Grip Strength

Perceived Recovery Status Scale

10	Very well recovered / Highly energetic	Expect Improved Performance
9		
8	Well recovered / Somewhat energetic	
7		
6	Moderately recovered	Expect Similar Performance
5	Adequately recovered	
4	Somewhat recovered	
3		
2	Not well recovered / Somewhat tired	Expect Declined Performance
1		
0	Very poorly recovered / Extremely tired	

Recovery Durations

During the workout
- Rest times
- Intra workout nutrition
- Volume autoregulation

Between workouts
- Nutrition
- Sleep
- Relaxation

Between Training weeks
- Light sessions
- Rest Days
- Massage

Between Training Phases
- Deloads
- Active Rest

Recovery Strategies

1. Manage Training Variables
2. Passive Recovery Strategies (sleep, stress management, life)
3. Active Recovery Strategies (deload, light sessions, tapers)
4. Nutrition (calories, macros, nutrient timing, food selection, hydration)
5. PEDs and Supplements
6. Rehab Modalities (Cold, Heat, Massage, EMS)

References

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